

Erica Sundermeyer

Physics PhD Student

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EDUCATION

Montana State University, Bozeman, MT

Fall 2026-Present

PhD in Physics

Advisor: Prof. Neil Cornish

University of Rochester, Rochester, NY

Spring 2026

BS in Physics and Astronomy; GPA: 3.93

Minors in Mathematics and Spanish

Advisor: Prof. Dominique Segura-Cox

HONORS, SCHOLARSHIPS

Montana State University University Fellowship (2026)

Merit-based award for outstanding new doctoral students

Astronaut Scholarship Foundation Award (2025)

Merit scholarship program with mission to ensure the United States maintains its leadership in science and technology by supporting exceptional college students pursuing degrees in STEM.

Univ. Rochester Dean's List (Fall 2022 - Spring 2026)

Univ. Rochester Whipple Science & Research Scholarship (Fall 2022 - Spring 2026)

Merit scholarship awarded upon admission to University of Rochester.

RESEARCH EXPERIENCE

University of Rochester, Department of Physics and Astronomy

Undergraduate researcher; Fall 2024 - Spring 2026

Advisor: Dr. Dominique Segura-Cox; d.segura-cox@rochester.edu

Observational radio astronomy project studying how evolution of star forming systems is affected by their larger environments, by analyzing the structure and kinematics of material surrounding young protostar Per-emb-2.

- Developed new method for obtaining angular momentum profile of asymmetric infalling streamer that does not rely on assumptions of axisymmetry. Verified method with theoretical streamline model.
- Developed research project and written proposal for Summer 2025.

Bachelor's Thesis: "First Specific Angular Momentum Analysis of an Infalling Streamer in a Class 0 Protostellar System"

Louisiana State University Physics and Astronomy REU

Undergraduate researcher; Summer 2024

Advisor: Prof. Gabriela Gonzalez; gonzales@lsu.edu

Performed bicoherence statistical study aimed at eliminating low frequency noise in LIGO (Laser Interferometer Gravitational-Wave Observatory) Livingston, using data

from LIGO 4th observing run. Identified coupling between noise signals in an auxiliary channel and main gravitational wave channel.

- Extrapolated analysis beyond initial scope of the project to the identification of coupling of noise signals at different frequencies in any pair of channels.
- Worked mostly independently, with weekly meetings with advisor and graduate student.
- Participated in professional development workshops (scientific communication, abstract writing, CV development, etc.) and assisted with LSU Physics and Astronomy department outreach events.

PRESENTATIONS

Innovators Symposium, Astronaut Scholarship Foundation, Houston, TX. August 14, 2026. Sundermeyer, E; “Astronomy on Tap Rochester, NY: Connecting Science with Community” (poster).

Innovators Symposium, Astronaut Scholarship Foundation, Houston, TX. August 16, 2025. Sundermeyer, E; “Radio Data Illuminates Physics of Early Protoplanetary Disk Formation” (talk).

Rochester Symposium for Physics Students, City College of New York, New York, NY. March 1, 2025. Sundermeyer, E; Segura-Cox, D; et. al.; “Streamer Rules Disk Formation in Young Binary System” (poster).

Summer Undergraduate Research Forum, Louisiana State University, Baton Rouge, LA. August 2 2024. Sundermeyer, E; Nandi, D; Gonzales, G; “Mystery Noise in Advanced LIGO Livingston Gravitational Wave Detector” (poster).

TEACHING

Montana State University, Department of Physics

Graduate Teaching Assistant

Fall 2026: Physics I (Calculus-based); College Physics II (Algebra-based)

University of Rochester, Department of Mechanical Engineering

Undergraduate Teaching Assistant

Fall 2025: Applied Boundary Value Problems

Upper-level partial differential equations course. Grading, weekly workshops, weekly office hours. Students: 52

University of Rochester, Department of Physics and Astronomy

Undergraduate Teaching Assistant

Fall 2025: Advanced Electricity and Magnetism I

Grading, weekly workshops and office hours. Students: 22

Spring 2025: Mechanics

Calculus-based introductory physics. Grading, weekly workshops, exam proctoring. Students: 15

Spring 2024: General Physics I

Algebra-based introductory physics. Weekly workshops and office hours. Students: 7

Fall 2023: Elementary Astronomy

Honors introductory astronomy. Workshops, office hours, grading, exam review sessions, supervision of observing nights with CEK Mees Observatory. Students: 26

COMMUNITY LEADERSHIP

Astronomy on Tap Rochester

Lead Organizer, September 2024 - May 2026

Re-established Astronomy on Tap satellite location in Rochester, NY. Organizing astronomy outreach events consisting of informal talks given by local experts and astronomy-themed trivia.

- Initiated communication with Astronomy on Tap HQ to discuss formation of Rochester satellite.
- Leading promotion and advertising efforts including logo design, social media presence, paper flyers, and communication with organizers of related events in the Rochester area.
- Coordinating with undergraduates, graduate students, postdocs, and faculty from Univ. of Rochester and Rochester Institute of Technology to recruit monthly speakers.

University of Rochester, Office of Undergraduate Research

Undergraduate Research Ambassador, Fall 2025 - Spring 2026

Volunteer duties including discussing research experience during open houses and info sessions, tabling events, and supporting students looking to get involved in research.

University of Rochester River Campus Libraries, Physics/Optics/Astronomy Library

Desk Assistant, Summer 2025 - Spring 2026

Fostered a welcoming environment in Physics/Optics/Astronomy Library as a community space for students. Shelves, organized, and maintained books, periodicals, and multimedia resources according to Library of Congress classification.

University of Rochester, Office of Orientation

Student Orientation Staff

Supervisor: Derek Pooley; orientation@rochester.edu

August 2025, August 2024: Welcome Week Leader

Managed setup and execution of orientation activities for incoming students; assisted with training of welcome week volunteers.

August 2023: Welcome Week Volunteer

University of Rochester, Office for Residential Life & Housing Services

First-Year Fellow, Fall 2023 - Spring 2024

Served as upper-class mentor to 61 students in first-year residence hall.

Membership

Sigma Pi Sigma: Physics & Astronomy Honor Society, U. of Rochester (2025 - 2026)

American Astronomical Society (Spring 2025 - Present)

American Physical Society (Fall 2024 - Present)

LIGO Scientific Collaboration (Summer 2024 - Spring 2025)

Society of Physics Students, University of Rochester (2022 - 2026)

Society of Women in Astronomy and Physics, University of Rochester (2022 - 2026)

SKILLS

Computational: Python; some experience with Git, Mathematica

Languages: English (fluent), Spanish (conversational)